

Proposition 1: Regret Lower Bound for Global Expert Selection

2026-06-27: Reconstructed proof replacing the original weak version.

1 Overview

The original Proposition 1 (全局专家排序不足) established the trivial fact that no single expert is simultaneously optimal across all states when risk functions cross. This observation is necessary for SCX but not sufficient as a theoretical foundation. We replace it with a **regret lower bound** theorem: any fixed global expert incurs a quantifiable penalty relative to the state-conditional optimal choice, and this penalty is bounded from below by a linear function of the number of states and the minimum risk gap between competing experts.

2 Notation and Setup

Symbol	Meaning
$\mathcal{S} = \{s_1, \dots, s_K\}$	State partition of the input space
$P(s) > 0$	Probability mass of state s ,
	$\sum_{s \in \mathcal{S}} P(s) = 1$
$\mathcal{M} = \{1, \dots, M\}$	Expert indices
$R_m(s)$	Expected risk of expert m in state s

Symbol	Meaning
$\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$	Loss function, assumed bounded: $\ell \in [0, L_{\max}]$

Assumption 1 (State partition). The state space $\mathcal{S}$ is a finite measurable partition of the input space $\mathcal{X}$ induced by a map $\Pi : \mathcal{X} \rightarrow \mathcal{S}$. Within each state s , the conditional distribution $P_{X|s}$ governs the input-output relationship.

Assumption 2 (Risk identifiability). For each state s and each expert m , the risk $R_m(s) = \mathbb{E}_{x \sim P_{X|s}}[\ell(f_m(x), f^*(x))]$ exists and is finite. Without loss of generality, all risks are bounded to $[0, L_{\max}]$.

3 Definitions

Definition 1 (Per-state expert ordering). For state s , order the experts by increasing risk:

$$R_{(1)}(s) \leq R_{(2)}(s) \leq \dots \leq R_{(M)}(s)$$

where $R_{(1)}(s) = \min_m R_m(s)$ is the best achievable risk in state s and $R_{(2)}(s)$ is the second-best.

Definition 2 (State gap). The gap in state s is:

$$\delta(s) = R_{(2)}(s) - R_{(1)}(s) \geq 0$$

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The minimum gap across all states is:

$$\delta_{\min} = \min_{s \in \mathcal{S}} \delta(s)$$

Definition 3 (Global expert regret). For a fixed global expert m^* , its regret relative to the state-conditional optimal choice is:

$$\text{Regret}(m^*) = \sum_{s \in \mathcal{S}} P(s) \cdot [R_{m^*}(s) - \min_{m \in \mathcal{M}} R_m(s)]$$

This measures the excess risk incurred by committing to a single expert across all states, compared to the oracle that selects the best expert per state.

4 Theorem 1: Regret Lower Bound

4.1 Statement

Let $\mathcal{S}$ be a state partition with K states, and let $\delta_{\min}$ be the minimum gap as defined above. For any fixed global expert $m^* \in \mathcal{M}$, define its suboptimal state set:

$$\mathcal{S}_{\text{bad}}(m^*) = \{s \in \mathcal{S} : m^* \notin \arg \min_m R_m(s)\}$$

Then:

$$\text{Regret}(m^*) \geq \sum_{s \in \mathcal{S}_{\text{bad}}(m^*)} P(s) \cdot \delta(s) \geq P(\mathcal{S}_{\text{bad}}(m^*)) \cdot \delta_{\min}$$

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where $P(\mathcal{S}_{\text{bad}}(m^*)) = \sum_{s \in \mathcal{S}_{\text{bad}}(m^*)} P(s)$ is the total probability mass of the states where m^* is suboptimal.

4.2 Proof

Step 1. Fix m^* and consider any state s . If $s \in \mathcal{S}_{\text{bad}}(m^*)$, then m^* is not the optimal expert in state s . Therefore:

$$R_{m^*}(s) \geq R_{(2)}(s)$$

since the best possible risk in state s is $R_{(1)}(s)$ and the next best is $R_{(2)}(s)$, and m^* is worse than at least the best expert.

Step 2. The per-state regret contribution for $s \in \mathcal{S}_{\text{bad}}(m^*)$ is:

$$R_{m^*}(s) - \min_m R_m(s) = R_{m^*}(s) - R_{(1)}(s) \geq R_{(2)}(s) - R_{(1)}(s) = \delta(s)$$

The inequality follows from Step 1: $R_{m^*}(s) \geq R_{(2)}(s)$, and $\min_m R_m(s) = R_{(1)}(s)$.

Step 3. For $s \notin \mathcal{S}_{\text{bad}}(m^*)$, the regret contribution is zero by definition since m^* is optimal in that state. Summing over all states:

$$\begin{aligned}
\text{Regret}(m^*) &= \sum_{s \in \mathcal{S}} P(s) \cdot [R_{m^*}(s) - \min_m R_m(s)] \\
&= \sum_{s \in \mathcal{S}_{\text{bad}}(m^*)} P(s) \cdot [R_{m^*}(s) - \min_m R_m(s)] \quad (\text{zero contribution elsewhere}) \\
&\geq \sum_{s \in \mathcal{S}_{\text{bad}}(m^*)} P(s) \cdot \delta(s) \quad (\text{by Step 2}) \\
&\geq \delta_{\min} \cdot \sum_{s \in \mathcal{S}_{\text{bad}}(m^*)} P(s) \quad (\text{since } \delta(s) \geq \delta_{\min}) \\
&= P(\mathcal{S}_{\text{bad}}(m^*)) \cdot \delta_{\min}
\end{aligned}$$

This completes the proof. $\square$

4.3 Tightness

The bound is tight. Consider a two-state, two-expert system with $P(s_1) = P(s_2) = 0.5$, $R_1(s_1) = 0$, $R_2(s_1) = \delta$, $R_1(s_2) = \delta$, $R_2(s_2) = 0$. Then $\delta_{\min} = \delta$, and for $m^* = 1$:

$$\text{Regret}(1) = P(s_2) \cdot \delta = 0.5\delta = P(\mathcal{S}_{\text{bad}}(1)) \cdot \delta_{\min}$$

The bound is attained exactly. For $m^* = 2$, symmetry gives the same value.

5 Corollary: Risk Crossing Implies Positive Regret

5.1 Statement

If there exist two states $s_1, s_2 \in \mathcal{S}$ and two experts $a, b \in \mathcal{M}$ such that:

$$R_a(s_1) < R_b(s_1) \quad \text{and} \quad R_a(s_2) > R_b(s_2)$$

(i.e., the expert risk functions **cross**), then for any global expert m^* :

$$\text{Regret}(m^*) \geq \delta_{\min} \cdot \min\{P(s_1), P(s_2)\} > 0$$

In particular, when the crossing involves the optimal experts (i.e., a is optimal in s_1 and b is optimal in s_2), no single expert can achieve zero regret.

5.2 Proof

Under the crossing condition, in state s_1 the best expert is at least as good as a (since $R_a(s_1) < R_b(s_1)$), and in state s_2 the best expert is at least as good as b . No single expert m^* can be simultaneously equal to both a and b (since $a \neq b$ by the crossing condition), so at least one of s_1 or s_2 is in $\mathcal{S}_{\text{bad}}(m^*)$. Therefore $P(\mathcal{S}_{\text{bad}}(m^*)) \geq \min\{P(s_1), P(s_2)\}$, and applying Theorem 1 gives:

$$\text{Regret}(m^*) \geq \delta_{\min} \cdot \min\{P(s_1), P(s_2)\} > 0$$

The strict positivity follows because $P(s_1), P(s_2) > 0$ (states are assumed to have positive probability) and $\delta_{\min} > 0$ when crossing involves the optimal experts (since $\delta(s) > 0$ if the optimal expert in state s is strictly better than all others). $\square$

6 Corollary: Co-monotonicity and Zero Regret

6.1 Definition (Co-monotonicity)

A family of risk functions $\{R_m\}_{m=1}^M$ is **co-monotonic** across states if for any two states $s_1, s_2 \in \mathcal{S}$ and any two experts $a, b \in \mathcal{M}$:

$$(R_a(s_1) - R_b(s_1)) \cdot (R_a(s_2) - R_b(s_2)) \geq 0$$

Equivalently, the ordering of experts by risk is invariant across states: there exists a total order $\preceq$ on $\mathcal{M}$ such that $R_a(s) \leq R_b(s)$ for all $s \in \mathcal{S}$ whenever $a \preceq b$.

6.2 Statement

Let m_{opt} be the best global expert:

$$m_{\text{opt}} = \arg \min_m \text{Regret}(m)$$

Then $\text{Regret}(m_{\text{opt}}) = 0$ **if and only if** there exists an expert m^* that is simultaneously optimal in every state:

$$\forall s \in \mathcal{S} : R_{m^*}(s) = \min_m R_m(s)$$

A **sufficient condition** for this is pairwise co-monotonicity of all expert risk functions: when experts are co-monotonic, the globally minimum-risk expert under the total order is uniformly optimal across all states, giving zero regret.

6.3 Proof

($\Leftarrow$). If an expert m^* is optimal in every state, then $\mathcal{S}_{\text{bad}}(m^*) = \emptyset$ and Theorem 1 gives $\text{Regret}(m^*) \geq 0$. Direct calculation gives $\text{Regret}(m^*) = 0$ since each term $R_{m^*}(s) - \min_m R_m(s) = 0$.

($\Rightarrow$). Suppose $\text{Regret}(m_{\text{opt}}) = 0$. Then for every state s , $P(s) \cdot [R_{m_{\text{opt}}}(s) - \min_m R_m(s)] = 0$. Since $P(s) > 0$, we must have $R_{m_{\text{opt}}}(s) = \min_m R_m(s)$ for all s , establishing that m_{opt} is simultaneously optimal everywhere.

Co-monotonicity as sufficient condition. Under co-monotonicity, the experts are totally ordered by risk. Let m^* be the minimal expert under this order (i.e., $R_{m^*}(s) \leq R_m(s)$ for all m and all s). Then m^* is optimal in every state, and by the ($\Leftarrow$) direction, $\text{Regret}(m^*) = 0$. $\square$

6.4 Testable Condition for Co-monotonicity

Given empirical risk estimates $\hat{R}_m(s)$ computed from data, co-monotonicity can be tested pairwise:

1. **Pairwise sign test.** For each pair (a, b) , compute the sign vector:

$$\sigma_{a,b}(s) = \text{sgn}(\hat{R}_a(s) - \hat{R}_b(s))$$

for all $s \in \mathcal{S}$. If for every pair $\sigma_{a,b}(s)$ is constant across s (all entries are $+1$, all -1 , or all 0), then the data are consistent with co-monotonicity.

2. **Rank correlation test.** For each pair (a, b) , compute Kendall's τ or Spearman's ρ of the risk vectors $(\hat{R}_a(s))_{s \in \mathcal{S}}$ and $(\hat{R}_b(s))_{s \in \mathcal{S}}$. Under co-monotonicity, all pairwise rank correlations equal $+1$. Deviations indicate risk crossing.

3. **Global crossing count.** The number of crossing pairs:

$$C = |\{(a, b, s_1, s_2) : (R_a(s_1) - R_b(s_1))(R_a(s_2) - R_b(s_2)) < 0\}|$$

quantifies the departure from co-monotonicity. When $C > 0$, regret must be positive for any global expert, and the lower bound of Theorem 1 applies.

7 Empirical Regret Bound

In practice, risks are estimated from finite data. Let $\hat{R}_m(s)$ be the empirical risk estimate based on n_s samples in state s .

7.1 Statement

With probability at least $1 - \eta$, for every expert m , the regret satisfies:

$$\text{Regret}(m) \geq \sum_{s \in \mathcal{S}} P(s) \cdot [\hat{R}_m(s) - \min_{m'} \hat{R}_{m'}(s)] - L_{\max} \sum_{s \in \mathcal{S}} P(s) \cdot \sqrt{\frac{2 \log(2M/\eta)}{n_s}}$$

7.2 Proof Sketch

By the Hoeffding inequality, for each state s and expert m :

$$|\hat{R}_m(s) - R_m(s)| \leq L_{\max} \sqrt{\frac{2 \log(2M/\eta)}{n_s}}$$

with probability $1 - \eta/M$. A union bound over M experts and K states, combined with the decomposition of regret, yields the result. This bound allows practitioners to compute a **high-probability lower bound** on regret from finite-sample risk estimates. $\square$

8 Implications for SCX

1. **Regret as a design principle.** Theorem 1 quantifies the penalty for ignoring state structure. The lower bound $P(\mathcal{S}_{\text{bad}}(m^*)) \cdot \delta_{\min}$ shows that regret grows linearly with both the mass of suboptimal states and the minimum risk gap.
 2. **State discovery drives regret reduction.** SCX’s state discovery directly reduces $P(\mathcal{S}_{\text{bad}}(m^*))$ by identifying regions where different experts excel. A perfect state partition (where each state has a single optimal expert) drives regret to zero for the state-conditioned system.
 3. **Co-monotonicity as a diagnostic.** The co-monotonicity test of Section 6.4 provides a data-driven criterion for whether a global expert is sufficient. When the test fails, SCX’s state-conditioned approach is guaranteed to improve over any fixed expert.
 4. **Sample-efficient testing.** The empirical bound of Section 7 enables practitioners to compute statistically valid lower bounds on regret with finite data, guarding against false conclusions due to estimation noise.
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References

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